



NORTH SYDNEY GIRLS HIGH SCHOOL

HSC MATHEMATICS ASSESSMENT TASK

TERM 2 – 2006

Time Allowed: 1 hour + 2 minutes reading time

Instructions:

- Start each question on a new page
- Write on one side of the paper only, work down the page and do not work in columns
- Leave a margin on the left hand side of the page
- Show all necessary working
- Marks may not be awarded for untidy or poorly arranged work
- Diagrams are not drawn to scale
- There are five questions
- Marks are as indicated

This task is worth 20% of the HSC Assessment Mark

Name: _____

Question 1 (7 marks)**Marks**

- a. Evaluate $\log_8 32$ 1
- b. Simplify $\log_5 45 + \log_5 40 - \log_5 72$ 1
- c. Given that $\log_2 3 = 1.58$, find $\log_2 12$ 1
- d. Find $\int_0^1 (e^x + 1) dx$ in exact form 2
- e. Evaluate $\int_0^3 \frac{x}{1+x^2} dx$ 2

Question 2 (10 marks) **Start a new page.**

- a. For the curve $y = 3 \sin 2x$
- State the period and amplitude. 2
 - Sketch the curve between $0 \leq x \leq 2\pi$ 1
- b. Find all solutions of $2 \sin x + 1 = 0$ in the interval $0 \leq x \leq 2\pi$.
(Give your answer/s as exact values). 2
- c. Differentiate the function $y = 3 \sin x + 4 \cos x$ 2
- d. Evaluate $\int_0^{\frac{\pi}{4}} (\sin 2x + \sec^2 x) dx$ 3

Question 3 (9 marks) **Start a new page.**

- a. The following table gives the values of $f(x) = x \log x$. Use Simpson's Rule with these 5 function values to find an approximate value of $\int_1^5 x \log x dx$.
(Give your answer to 2 decimal places) 3

x	1	2	3	4	5
$f(x)$	0	1.39	3.3	5.55	8.05

- b.
- Find the derivative of $\tan 2x$. 1
 - Find the volume of the solid formed when the curve $y = \sec 2x$ is rotated about the x -axis between the ordinates $x = 0$ and $x = \frac{\pi}{8}$. 2
- c.
- The x -coordinate of one point of intersection between the line $y = x + 1$ and the parabola $y = 2x^2$ is $x = -\frac{1}{2}$. Using quadratic equations, or otherwise, find the x -coordinate of the other point of intersection. 1
 - Hence, find the area enclosed between the line and the parabola. 2

Question 4 (9 marks) **Start a new page.****Marks**

- a. The area under the curve $y = \frac{1}{x}$ between $x = 1$ and $x = b$ is equal to 1 square unit.

What is the value of b ?

3

- b. A solid of revolution is formed by rotating about the x -axis the curve $y = 2(1 + e^{2x})$ between the ordinates $x = 0$ and $x = \frac{1}{2}$. Show the volume of the solid is

$$\pi(e^2 + 4e - 3) \text{ units}^3.$$

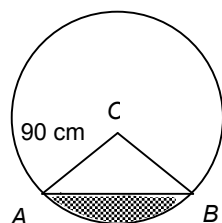
3

- c. For the curve $y = \log(1 + \sin x)$, show that the second derivative is given by

$$\frac{-1}{1 + \sin x}$$

3**Question 5** (8 marks) **Start a new page.**

- a. In the diagram the length of the arc AB is 50 cm. The radius of the circle is 90 cm.



- i. Find $\angle AOB$ in radians

1

- ii. Find the shaded area correct to the nearest cm^2 .

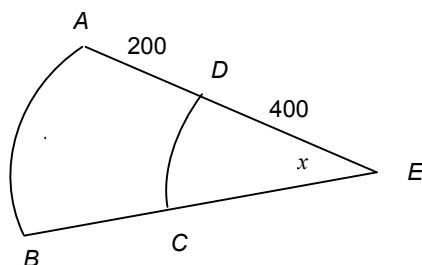
2

- b. In the figure AB and CD are circular arcs which subtend an angle x radians at the centre E , where $0 \leq x \leq \pi$. The length AD is 200 metres and DE is 400 metres.

- i. Find the length of each of the arcs AB and CD in terms of x .

2

- ii. A person lives at A and wants to walk to the train station at B . The paths AB , BC , CD and DA form a road. For what values of x is it shorter for the person to walk along the route $ADCB$ rather than along the arc AB ?

3**END OF TEST**

Solutions

$$(1) \quad (a) \quad \log_8 32 = x$$

$$\therefore 8^x = 32 = 2^5 \Rightarrow 2^{3x} = 2^5$$

$$\therefore x = \frac{5}{3}$$

Alternatively

$$\log_8 32 = \frac{\ln 32}{\ln 8} = \frac{5}{3} \quad (\text{calculator})$$

$$(b) \quad \log_5 45 + \log_5 40 - \log_5 72 = \log_5 \left(\frac{45 \times 40}{72} \right) = \log_5 25 = 2$$

$$(c) \quad \log_2 12 = \log_2 (2^2 \times 3) = \log_2 (2^2) + \log_2 (3) = 2 + \log_2 (3) \approx 3.58$$

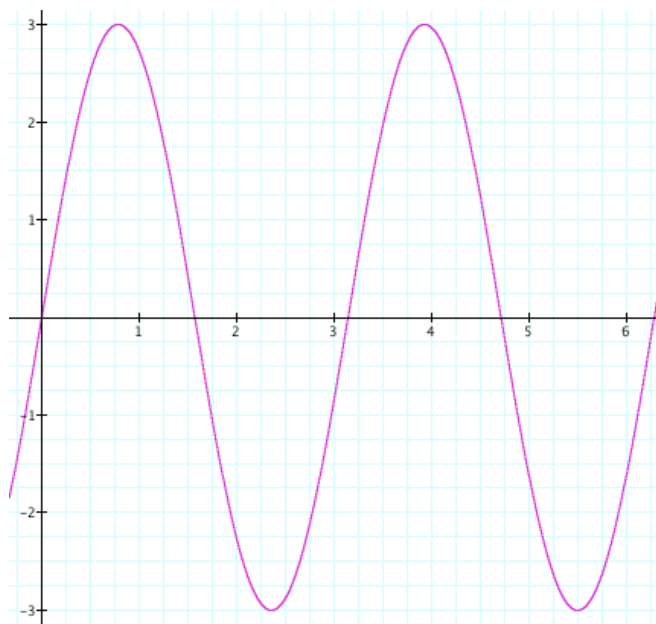
$$(d) \quad \int_0^1 (e^x + 1) dx = [e^x + x]_0^1 = (e + 1) - (1 + 0) = e$$

$$(e) \quad \int_0^3 \frac{x}{1+x^2} dx = \frac{1}{2} \int_0^3 \frac{2x}{1+x^2} dx = \frac{1}{2} [\ln(1+x^2)]_0^3 = \frac{1}{2} (\ln 10 - \ln 1) = \frac{1}{2} \ln 10$$

$$(2) \quad (a) \quad (i) \quad \text{Amplitude} = 3$$

$$\text{Period} = \frac{2\pi}{2} = \pi$$

(ii)



$$(b) \quad 2 \sin x + 1 = 0 \Rightarrow \sin x = -\frac{1}{2}$$

$$\therefore x = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$(c) \quad y = 3 \sin x + 4 \cos x$$

$$\therefore y' = 3 \cos x - 4 \sin x$$

$$(d) \quad \int_0^{\frac{\pi}{4}} (\sin 2x + \sec^2 x) dx = \left[-\frac{\cos 2x}{2} + \tan x \right]_0^{\frac{\pi}{4}} = (1) - \left(-\frac{1}{2} \right) = \frac{3}{2}$$

(3) (a)

x	1	2	3	4	5
$f(x)$	0	1.39	3.3	5.55	8.05
w	1	4	2	4	1
$w \times f(x)$	0	5.56	6.6	22.2	8.05

$$h = 1$$

$$\sum wf(x) = 42 \cdot 41 \Rightarrow \int_1^5 x \log x dx \approx \frac{h}{3} \times 42 \cdot 41 = 14 \cdot 14 \text{ (2dp)}$$

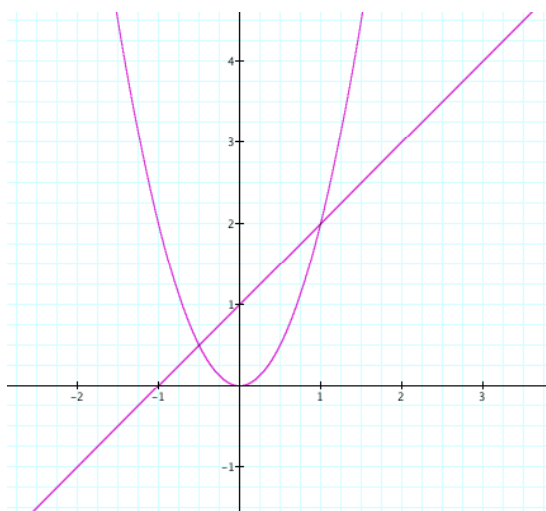
(b) (i) $\frac{d}{dx}(\tan 2x) = 2 \sec^2 2x$

(ii)
$$V = \pi \int_0^{\frac{\pi}{8}} \sec^2 2x dx = \frac{\pi}{2} \int_0^{\frac{\pi}{8}} 2 \sec^2 2x dx$$

$$= \frac{\pi}{2} [\tan 2x]_0^{\frac{\pi}{8}} = \frac{\pi}{2} \times 1$$

$$= \frac{\pi}{2} \text{ cu}$$

(c) (i)



Let $\alpha = -\frac{1}{2}$ and β be the x -coordinates of the points of intersection

$$y = x + 1 = 2x^2 \Rightarrow 2x^2 - x - 1 = 0$$

$$\therefore \alpha + \beta = \frac{1}{2} \Rightarrow -\frac{1}{2} + \beta = \frac{1}{2} \Rightarrow \beta = 1$$

(ii)
$$\text{Area} = \int_{-\frac{1}{2}}^1 (x + 1 - 2x^2) dx = \left[\frac{x^2}{2} + x - \frac{2}{3}x^3 \right]_{-\frac{1}{2}}^1 = 1\frac{1}{8}$$

$$(4) \quad (a) \quad \int_1^b \frac{dx}{x} = 1 \Rightarrow [\ln x]_1^b = 1$$

$$\therefore \ln b = 1 \Rightarrow b = e$$

$$(b) \quad V = \pi \int_0^{\frac{1}{2}} 4(1 + e^{2x})^2 dx = \pi \int_0^{\frac{1}{2}} (4 + 8e^{2x} + 4e^{4x}) dx$$

$$= \pi \left[4x + 4e^{2x} + e^{4x} \right]_0^{\frac{1}{2}}$$

$$= \pi [2 + 4e + e^2 - (4 + 1)]$$

$$= \pi [e^2 + 4e - 3] \text{ cu}$$

$$(c) \quad y = \ln(1 + \sin x)$$

$$y' = \frac{\cos x}{1 + \sin x}$$

$$y'' = \frac{(1 + \sin x) \times (-\sin x) - \cos x \times \cos x}{(1 + \sin x)^2}$$

$$= \frac{-\sin x - \sin^2 x - \cos^2 x}{(1 + \sin x)^2} = \frac{-\sin x - 1}{(1 + \sin x)^2}$$

$$= -\frac{1 + \sin x}{(1 + \sin x)^2}$$

$$= -\frac{1}{1 + \sin x}$$

$$(5) \quad (a) \quad (i) \quad 90 \times \angle AOB = 50$$

$$\therefore \angle AOB = \frac{5}{9}$$

$$(ii) \quad \text{Let } \theta = \angle AOB$$

$$\text{Shaded area} = \frac{1}{2} r^2 (\theta - \sin \theta) = \frac{1}{2} \times 90^2 \left(\frac{5}{9} - \sin \frac{5}{9} \right)$$

$$\approx 114 \text{ cm}^2$$

$$(b) \quad (i) \quad CD = 400x; \quad AB = 600x$$

$$(ii) \quad ADCB = 400 + 400x$$

$$ADCB < \text{arc } AB \Rightarrow 600x > 400 + 400x$$

$$\therefore 200x > 400$$

$$\therefore x > 2$$

So when the angle is greater than 2 radians (ie greater than $114^\circ 35'$), arc AB is the greater distance